



NAME: _____ ID: _____

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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circumf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

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Question 3. Revenge is colored (10% of total; only 12th grade) Day 1

3.1	3.2	Total	Total
12	28	40	10

The boys were very thoughtful about what happened in the previous question and agreed that the best way was to answer the girls, "with a little of their own medicine." Bárbaro, Alejandro *Murad*, Osmany, Brayan, Nathan, Pedro, and the two Carlos came up with a substance-identification question for the girls.

Again, the setting was the school's Chemistry laboratory, but this time, with the principal's warning not to "cause trouble". Each student showed a transparent flask containing a colorless aqueous solution of the inorganic substances: AgNO_3 , BaCl_2 , CuSO_4 , Na_2S , VOSO_4 and ZnSO_4 .

Murad and Pedro brought blue solutions. The samples of Bárbaro, Osmany, Brayan and Nathan were colorless. However, Osmany's had a characteristic odor and Brayan's, at the end of the demonstration, showed grayish particles when observed against the light. The students then conducted careful assays to provide clues for identification.

Carlos Manuel added a few drops of a sulfuric-acidulated KMnO_4 solution to test tubes containing each of the above substances. The solutions of Bárbaro and Murad were colored purple, with no other apparent change occurring. Brayan's and Nathan's solutions were colored purple and became turbid, with the effect much more pronounced in Nathan's sample. On the contrary, the solution of Osmany caused a yellowish-white turbidity and a very unpleasant odor. Pedro's solution turned green with the first drops of permanganate, then yellow and finally yellow-violet, when excess reagent was added.

Carlos Daniel used a concentrated ammonia solution for identification. The solution of Bárbaro soon formed a white precipitate that dissolved slightly upon the addition of excess reagent. Murad's solution produced a sky-blue precipitate, which partially dissolved in excess of the reactant, yielding an intense navy-blue color. The solutions of Osmany and Nathan did not undergo any obvious change. Brayan's solution immediately formed a white precipitate, which over time turned brown and dissolved in excess of the reactant, resulting in a colorless solution. Pedro's solution formed a blue-green precipitate that, with excess reagent and prolonged exposure to air, partially dissolved, coloring the solution light yellow. The yellow solution turned orange when a strong acid was added.

3.1. Help the girls on the team identify the substances that each of the boys had.

3.2. Write down the balanced chemical equations of all the chemical reactions described.

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Question 7. Back to eleventh Grade (12% of total; only 11th and 12th grades) Day 1

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
6	8	10	8	8	8	48	12

7.1. Estimate at what temperature the sea water freezes; given that the mean salinity is 3.5% in mass, the mean density is 1.025 g/mL. Suppose that the only marine salt is NaCl. $K_{cr}(H_2O) = 1.86\text{ }^{\circ}\text{C}\cdot\text{kg}\cdot\text{mol}^{-1}$

7.2. In blast furnaces, where metal is produced from the reaction of ore (oxide) with carbon, several equilibria coexist. Three possible reactions of **one mole** of dioxygen with carbon and carbon monoxide are known. Estimate at what temperature the reduction of the mineral with carbon, forming monoxide, is the most thermodynamically favored reaction among the possible.

Substance	O ₂	C	CO	CO ₂
ΔH_f°	0	0	-111	-394
S°	205	6	198	214

7.3. You want to prepare 1.00 L of aqueous phosphate buffer solution of pH = 7.5 and total molar concentration equal to 0.50 mol/L. An aqueous solution of commercial phosphoric acid (~15 mol/L) and solid sodium hydroxide is available for this purpose. Which are the two species of phosphorus that predominate at this pH value? What amounts do you need to mix of each reagent? For H₃PO₄ we know that pK_{a1} = 2.15, pK_{a2} = 7.20 and pK_{a3} = 12.35.

7.4. Most chemical elements form hydroxides and sulfides that are poorly soluble in water. In the 2+ oxidation state, the vast majority of sulfides are more insoluble than hydroxides. In the 3+ and 4+ oxidation states, on the other hand, some sulfides exist only in the solid phase and are rapidly hydrolyzed to form hydroxides in the presence of moisture; for example, this occurs for iron(III) and aluminum. Suppose you have an aqueous solution of MnCl₂ 10⁻⁵ mol/L. Estimate the concentration of sodium sulfide required to initiate precipitation. What product is formed? $K_{ps}(\text{Mn}(\text{OH})_2) = 2.0\cdot 10^{-13}$, $K_{ps}(\text{MnS}) = 3.0\cdot 10^{-10}$; for H₂S we know pK_{a1} ~ 7.0 and pK_{a2} ~ 13.0.

7.5. Calculate the potential of the silver-silver chloride reference electrode Ag(s)|AgCl(s)|KCl(sat.) with respect to the normal hydrogen electrode, knowing that $K_{ps}^{\circ}(\text{AgCl}) = 1.77\cdot 10^{-10}$, $E^{\circ}(\text{Ag}^+/\text{Ag}) = 0.7996\text{ V}$ and that a saturated solution of KCl contains approximately 4.81 mol/L of salt. Note that in such a concentrated solution, the ions interact strongly, and their "effective concentration" or activity is only a factor or fraction of 0.588 (mean ionic activity coefficient) with respect to the molar concentration.

7.6. A green dihydrogen production plant uses an electrolyzer consisting of 200 cells. Each cell operates at a current of 5.00 A and requires 1.85 V. The performance or faradaic efficiency of the process is 98.5%, that is, this is the fraction that participates in the desired electrochemical reaction. Calculate the mass of dihydrogen produced each hour and the energy consumption.

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Question 8. Ouro Preto (15% of total; 11th and 12th grades) Day 1

8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	8.10	8.11	8.12	8.13	Total	Total
2	8	4	4	2	2	4	3	8	2	9	2	10	60	15

Last year, Cuba participated in the Mendeleev International Olympiad for the second time in Belo Horizonte, Brazil. The “picadillo team” was made up of Juan Carlos (called *Yepe* by everyone), Roy Roger, Marco Antonio, Ernesto Alejandro and Professor Julieta, who is also president of the Cuban Society of Chemistry. The result obtained was very relevant, with a silver medal and three bronze medals. During the event, the team visited the city of Ouro Preto, notorious for gold mining in the late 17th and early 18th centuries. The teacher, who loves chemistry very much, took the opportunity to remind the students of the electrochemical properties of this element.

8.1. Ernesto was struck by the fact that gold is widely used in electronics due to its easy malleability, ductility and excellent electrical conductivity. Why do the triad Cu, Ag and Au have the highest conductivity of all the elements in the periodic table?

8.2. *Yepe* learned that the city of Ouro Preto received that name because the gold ore found in the area was obscured by a mixture of oxides (approximately 28% and 30%, by mass of oxygen) of a certain transition element. Can you find out which element it is and the formula of its oxide?

Julieta really likes jewelry and working with metals. As she is inquisitive, she found out from the organizers what the composition of the gold medals awarded during the event was and the details of the coating process, and she also received one golden medal as a gift.

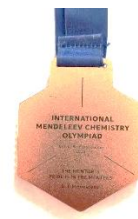
The medals were made of brass (Cu-Zn alloy), weighed approximately 210 g and had 5.0 μm of gold plating. The gold coating was performed in a solution of $\text{K}[\text{Au}(\text{CN})_2]$ 0.040 M, at pH = 4.0, using as stabilizer and buffer a solution of 75 g/L of citric acid (H_4Cy , 192 g/mol), at 50 °C, at a voltage of 2-4 V, with a current density of 0.50 A/dm² and a gold sacrificial anode. The faradaic efficiency of the process is close to 85%.

8.3. Gold(III) cyanide does not exist, but when the two ions come into contact in solution, a pseudohalogen is released, and a precipitate is formed. Formulate the reaction and explain why it happens.

8.4. Au(I) is unstable in solution, but can be stabilized by halide ions, cyanide, thiocyanate, and thiosulfate. Why? Calculate the disproportion constant of Au(I) in water at 25 °C.

Yepe wanted to know the exact composition of the medals. He determined that the mass of the medal that Julieta was given was 210 g and measured its dimensions, obtaining 40 mm side length and 6.0 mm thick. Using data available online, he was able to determine the composition but did not reveal it to his colleagues.

8.5. Assume for the rest of the question that the medal is shaped like a regular L-sided hexagon, with an area equal to $A = \frac{3\sqrt{3}}{2} \cdot L^2$. Calculate the approximate density of the alloy.



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8.6. The volume obtained by *Yepe* is approximate, because the medal was actually irregularly shaped. Explain a more accurate method of measuring volume in the laboratory.

8.7. The efficiency of the gold coating process is often affected because collateral half-reactions occur at the anode. One reaction releases gas and another reaction forms a precipitate that passivates the electrode; thus, requiring higher operating voltage. Formulate the balanced equations of these two reactions.

Roy then said that his father is a jeweler and that, in his workshop, he uses aqua regia (a mixture of HNO_3 and HCl , in 1:3 proportion) to dissolve certain pieces, because acids alone cannot do so. Julieta explained that this reaction is possible due to the formation of the complex $[\text{AuCl}_4]^-$.

8.8. Help Roy calculate the standard Au^{3+}/Au reduction potential in the absence of chlorides.

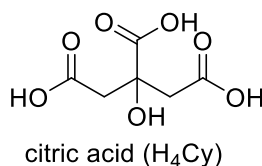
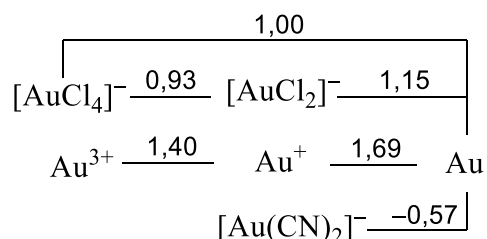
8.9. Formulate the possible reactions of gold with H^+ , NO_3^- and O_2 in the air, in the presence of chlorides. Determine which is thermodynamically favored, assuming a medium with $\text{pH} = 0$ and 1.0 M chloride.

8.10. In practice, the reaction you found above is not the one that happens. Propose an explanation.

8.11. When Marco calculated the amount of gold (19.3 g/cm^3) in grams consumed in a medal, he was surprised by the result. How long does it take to coat a gold medal?

8.12. At the end of the tour, the teacher, a specialist in organic compounds, asked the students why one of citric acid's constants is extremely small. What would you have answered?

8.13. A 2.00 g sample of filings from a gold medal was dissolved in hot nitric acid. The concentrated solution was cooled, filtered and neutralized with care to avoid precipitation of the metal hydroxides. Then, it was diluted until it completed a liter of solution in a volumetric flask. A 10 mL aliquot was transferred to an Erlenmeyer bottle, and a small amount of distilled water and 1 g of potassium iodide were added. The resulting suspension was quickly titrated with a 0.0100 M thiosulfate standard, using starch as an indicator. The volume of titrant required to achieve the first blue coloration was 21.1 mL. What is the total composition (in percent by mass) of the IMChO gold medal?



$K_{\text{ps}}(\text{AuCN}) = 1,0 \cdot 10^{-38}$ $E^\circ(\text{V}): \text{O}_2/\text{H}_2\text{O} = 1,23; \text{NO}_3^-/\text{NO} = 0,96$

$\text{pK}_a(\text{H}_4\text{Cy}) = 3,13; 4,76; 6,40; 14,4$ Air: 79% N_2 and 21% O_2 (v/v), 1,0 atm

Note: The author declares that no IMChO medal was damaged during the preparation of this question.

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Question 9. Alloys (8% of total; only 12th grade) Day 1

9.1	9.2	9.3	9.4	9.5	9.6	Total	Total
2	6	4	4	12	4	32	8

Metals can be mixed in different proportions to form alloys, creating new materials. Amalgams are alloys of mercury and are probably among the best known.

9.1. Sodium and zinc amalgam are used in Julia-Lythgoe olefination and Clemmensen reduction, respectively. What is the function of mercury, and why is this specific metal used?

The transport of mercury, metallic or in the form of salt, is prohibited on airplanes, because it can cause severe damage to the aluminum structure of the aircraft. The mercury amalgam, Al(Hg), that forms, catalyzes the oxidation of mercury by moisture in the air.

9.2. Why is aluminum inert under normal conditions? Write the equations of the reactions that occur in the presence of mercury, metallic and in the form of salt.

When highly electropositive metals are combined with highly electronegative metals or metalloids, compounds with a high ionic character are created, such as CsAu that melts at 590 °C and the Zintl phases NaTl and CaSi₂, in which the cations are intercalated in a lattice formed by the atoms of the most electronegative element.

9.3. Suggest what kind of three-dimensional structure the species Tl⁻ and Si²⁻ form in the Zintl phases.

9.4. Construct the molecular orbital diagram for the AuCs. Show only the valence orbitals.

Two metals **A** and **B** from the same group of the periodic table can be mixed in different proportions. When the combination is equimolecular, the mass fraction of **A** is 37%, and a liquid compound is obtained. That liquid is used as a coolant in nuclear reactors for its high thermal conductivity. At low temperatures, it is possible to obtain a solid compound with 54% **A** in mass. Both metals and their compounds explode when they come into contact with water.

9.5. Find the identity of metals **A** and **B**, and the formula for the intermetallic compounds.

9.6. Write the balanced equation of the reaction of the equimolecular compound between metals **A** and **B** with excess of: i) water and ii) oxygen.

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Question 10. Marginalized elements (15% of total; only **12th grade**) **Day 1**

To the memory of Tec. Nicasio *Kino* Joaquín Borrego, FQ-UH Organic Department

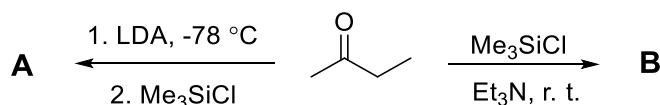
10.1	10.2	10.3	10.4	10.5	10.6	10.7	10.8	Total	Total
6	12	18	2	6	2	4	10	60	15

Often, novice students prefer to focus on the organic chemistry of C, H, N, O compounds and are very afraid of molecules with B, P, S, and Si, the four horsemen of the organic chemical apocalypse. This question is meant to unleash their worst fears. But if you have basic knowledge of structure and reactivity, persistence and a little imagination, you will surely be able to answer it.

To answer the following questions, consider the properties of the elements in question.

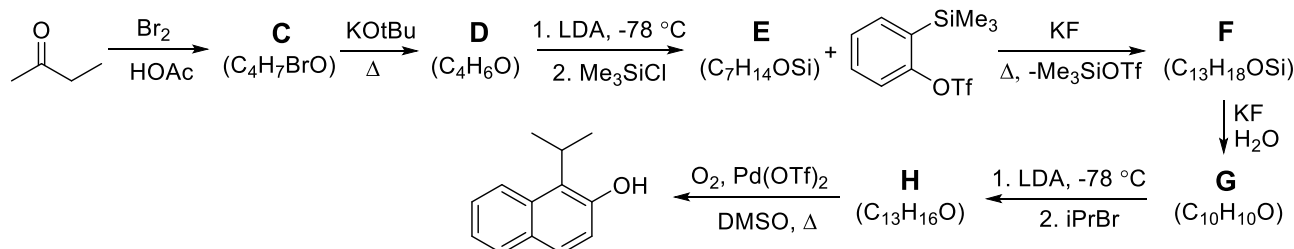
1. Sulfur compounds have various oxidation states; they can be nucleophilic or electrophilic and stabilize cations and anions. The S atom is soft and can be removed by reduction or oxidation.
2. Boron compounds are electrophilic due to the vacant p-orbital. The B atom has a high affinity for oxygen and can be eliminated by oxidation.
3. Silicon is more electropositive than carbon so that it can stabilize β -carbocations; it is a very hard atom, which has a high affinity for fluorine and oxygen.
4. Phosphorus compounds are restricted to oxidation states 3+ and 5+; they can be nucleophilic or electrophilic. The P atom is soft but has a high affinity for oxygen.

It is possible to control the regioselectivity of enolate formation or its synthetic equivalents by adjusting reaction conditions. Consider the reactions of butanone that form silyl enol ethers.



10.1. Represent the structures of isomeric compounds **A** and **B**. Explain the origin of the selectivity.

10.2. Consider the following synthesis of a β -naphthol derivative and represent the structures of **C-H**.



Phosphorus reagents are frequently used to exchange functional groups in organic reactions. For example, in the reactions of Apple, Corey-Fuch and Mitsunobu.

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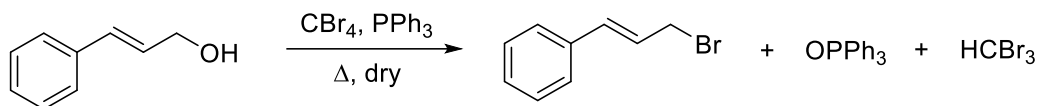
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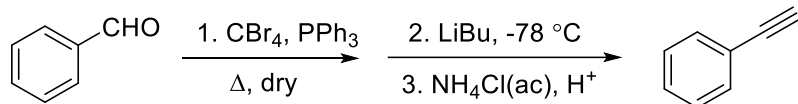
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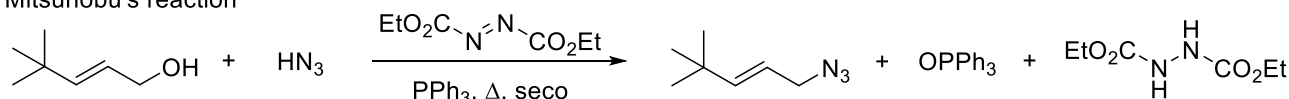
Appel's reaction



Corey-Fuch's reaction



Mitsunobu's reaction



10.3. Propose a mechanism for each of the above reactions. Indicate the movement of the electrons. Why is the use of anhydrous conditions necessary?

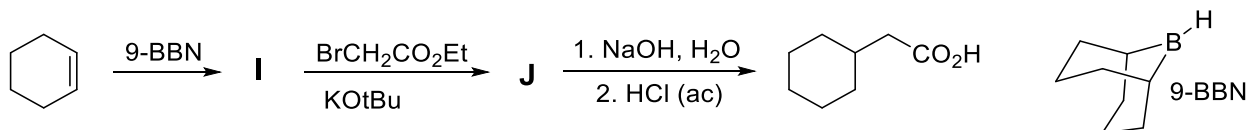
10.4. Propose other alternative reagents to transform the allyl alcohol into the corresponding azide.

The reaction of sodium hydride with boron trichloride in the presence of ether allows the obtaining of two well-known reducing agents **X** and **Y**, depending on the amounts of the reactants. **X** is a molecular gas that rapidly tends to dimerize to **X₂** and must be stored as dry ether solution, while **Y** is a relatively stable ionic salt that explodes on contact with water.

10.5. Identify substances **X** and **Y** and write the equations for obtaining them.

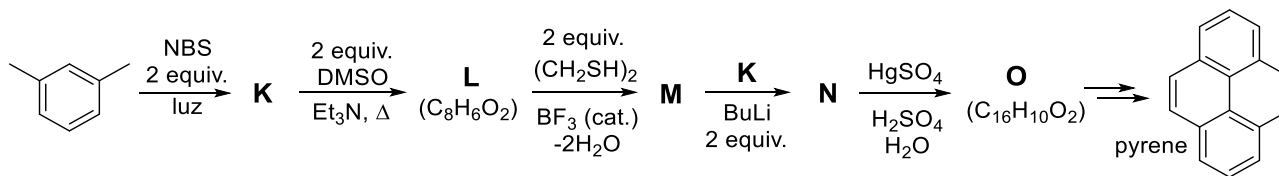
10.6. **Y** is a nucleophilic reductant, which reduces aldehydes, ketones, imines and esters (under heating). In contrast, **X** is an electrophilic reductant, which reduces alkenes, alkynes, carboxylic acids, and amides. Explain the different reactivity between these substances.

Boron compounds can be used to synthesize ketones and substituted esters via carbanions, as illustrated in the example.



10.7. Represent the structures of compounds **I** and **J**.

Thiols allow aldehydes (electrophiles) to be masked in the form of dithians (nucleophiles) and to reverse the polarity (umpolung) of chemical reactions. Below you find a possible synthesis of pyrene.



10.8. Represent the structures of **K-O** compounds. NBS: N-bromosuccinimide.

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